Quantification: I534_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	32
interior lacuna count	25
frame-edge lacuna count	7

Interior lacunae only: mean, median and sample SD (n = 25)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2817.72	1943	2635.45	px^2
major_axis_length_px	lacuna shape	83.2391	74.781	32.4447	px
minor_axis_length_px	lacuna shape	44.1848	37.2098	22.4578	px
aspect_ratio	lacuna shape	2.0619	2.0239	0.6289	unitless
eccentricity	lacuna shape	0.8262	0.8694	0.1152	unitless
solidity	lacuna shape	0.7461	0.7383	0.0965	unitless
roots_count	headline	10.72	9	5.8632	count
ring_length_r30_px	headline	288.48	274	140.165	px
ring_length_r60_px	reported	548.28	483	294.834	px
owned_length_px	ownership-dependent	664.791	431.333	606.178	px
edge_count	ownership-dependent	33	23	29.199	count
mean_edge_length_px	ownership-dependent	20.2836	19.6399	6.0389	px
perimeter_px	normalised	273.398	253.279	124.13	px
ring_area_r30_px2	normalised	8925.8	7690	3535.79	px^2
ring_area_r60_px2	normalised	17509.3	16187	7520.57	px^2
in_frame_fraction_r30	normalised	0.987	1	0.0329	unitless
in_frame_fraction_r60	normalised	0.9657	1	0.0725	unitless
ring_density_r30	normalised	0.0325532	0.0315741	0.0109642	px^-1
ring_density_r60	normalised	0.0314602	0.0342956	0.0109076	px^-1
roots_per_100px_perimeter	normalised	3.8988	3.9482	1.0904	per 100 px
ring_attached_length_r30_px	network v2	262.64	232	142.636	px
ring_attached_length_r60_px	network v2	438.16	348	294.951	px
ring_length_w_r30_px	network v2	327.358	309.35	158.823	px
ring_length_w_r60_px	network v2	621.849	549.902	335.246	px
sholl_crossings_r10	network v2	6.56	6	3.4409	count
sholl_crossings_r20	network v2	7.32	6	4.0386	count
sholl_crossings_r30	network v2	7.96	7	4.5321	count
loop_count	loops, report only	1.68	0	2.7343	count
loop_edge_share	loops, report only	0.1102	0	0.1462	share
ring_width_mean_r30_px	width	7.8747	7.7108	1.2345	px

Per field
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measure	value	units
analysed_area_px2	887774	px^2
total_skeleton_length_px	25070	px
canalicular_length_density_per_px	0.0282392	px^-1
junction_count	548	count
junction_density_per_px2	0.00061727	px^-2
lacuna_count	32	count

lacunae_per_px2	3.605e-05	px^-2
skeleton_component_count	310	count
mean_component_length_px	80.871	px
median_component_length_px	22.5	px
field_length_density_w_per_px	0.0320322	px^-1
field_density_without_flagged_per_px	0.0296288	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	184	px
unattached_component_count	12	count
canalicular_length_density_attached_per_px	0.0282392	px^-1
field_length_density_with_unattached_per_px	0.0284464	px^-1
loop_count	58	count
loop_edge_share	0.15	share
edge_count	1387	count

Canalicular width, the new measure

measure	value	units
width_mean_px	8.2827	px
width_median_px	8.2462	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	23469	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	5018	113.612	66.0072	1.7212	0.8139	0.7134	-1.5671	33.3908	243.731	True
2	1429	75.9455	34.6321	2.1929	0.89	0.5788	0.9821	47.7124	162.336	False
3	12478	404.05	54.0339	7.4777	0.991	0.6138	1.5354	38.8529	844.287	True
4	1943	78.2261	36.7629	2.1279	0.8827	0.7692	-1.4856	56.1004	636.36	False
5	1748	63.7004	52.4126	1.2154	0.5683	0.625	1.3536	36.7677	399.808	False
6	2001	74.781	36.9486	2.0239	0.8694	0.794	-1.1924	101.66	153.378	False
7	4592	90.8676	67.3599	1.349	0.6712	0.8423	-1.2953	137.617	765.059	False
8	902	70.1443	23.2471	3.0173	0.9435	0.6677	-1.2472	150.552	535.413	False
9	18263	251.192	114.111	2.2013	0.8909	0.7282	-1.0667	218.389	97.7195	True
10	2333	67.9441	45.1639	1.5044	0.7471	0.9554	-0.4874	234.38	1000.91	True
11	11887	168.982	95.145	1.7761	0.8264	0.8606	-1.2985	286.803	681.158	False
12	697	50.8222	20.9969	2.4205	0.9107	0.7383	1.293	287.073	212.017	False
13	3532	87.4182	55.1855	1.5841	0.7756	0.7638	1.235	329.842	344.026	False
14	4129	107.816	50.0459	2.1543	0.8857	0.9077	-1.34	378.661	498.241	False
15	889	60.4502	27.413	2.2052	0.8913	0.6247	-1.2388	363.547	49.8526	False
16	1735	68.9793	37.2098	1.8538	0.842	0.6893	-0.903	463.884	594.474	False
17	751	42.1527	25.0435	1.6832	0.8044	0.8632	-1.1817	459.581	482.401	False
18	5417	114.374	70.6956	1.6178	0.7861	0.6168	1.316	527.301	55.7192	True
19	3037	99.839	58.2931	1.7127	0.8118	0.571	1.4007	558.019	301.395	False
20	4073	115.2	49.8465	2.3111	0.9015	0.7391	0.3504	603.966	867.145	False
21	2337	72.0848	52.1015	1.3835	0.6911	0.6974	-1.4357	607.677	651.435	False
22	1679	74.0209	37.2141	1.9891	0.8644	0.724	-0.9638	634.934	701.784	False
23	7591	130.597	95.8386	1.3627	0.6793	0.6789	-0.2363	683.477	82.1523	False
24	4140	142.999	43.9592	3.253	0.9516	0.7236	-1.1238	670.561	384.423	False

25	554	30.7716	24.9383	1.2339	0.5858	0.895	1.0673	729.574	697.603	False
26	549	48.7614	15.8873	3.0692	0.9454	0.8356	-0.9101	738.328	786.444	False
27	640	57.661	17.8443	3.2313	0.9509	0.7018	-0.3969	793.93	358.258	False
28	2036	78.7104	35.4441	2.2207	0.8929	0.8879	-1.002	828.682	94.3777	False
29	45573	496.877	154.577	3.2144	0.9504	0.7075	1.5056	966.648	733.764	True
30	1332	74.3708	26.5147	2.8049	0.9343	0.75	-1.2358	829.074	388.601	False
31	6240	115.676	84.3355	1.3716	0.6844	0.7226	-1.0938	881.946	274.251	False
32	1277	51.634	38.2716	1.3491	0.6713	0.7854	1.2894	872.15	1003.8	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	15	337	427	568.174	29	19.5922	6.7223
2	12	274	358	364.877	23	15.8642	6.4715
3	24	502	617	535.973	32	16.7491	6.8655
4	10	293	572	426.534	28	15.2334	7.208
5	9	352	597	848.37	50	16.9674	7.3539
6	7	240	321	161.439	10	16.1439	6.8457
7	15	432	634	749.227	38	19.7165	7.593
8	8	322	799	783.512	40	19.5878	7.7452
9	18	515	788	919.666	48	19.1597	7.5529
10	6	155	358	401.421	19	21.1274	8.7355
11	26	659	1235	2350.11	115	20.4357	8.1509
12	7	204	390	431.333	23	18.7536	8.7157
13	16	500	1055	2184.78	102	21.4194	8.5149
14	11	341	658	1054.25	55	19.1681	8.3777
15	7	179	235	246.492	10	24.6492	6.1524
16	12	360	668	1001.49	35	28.614	7.7108
17	5	155	258	175.125	10	17.5125	8.2917
18	19	351	508	725.027	34	21.3243	5.9354
19	19	385	862	1396.18	70	19.9455	7.8658
20	14	408	1030	1085.1	51	21.2766	7.1052
21	15	277	585	667.245	36	18.5346	7.6247
22	9	167	339	207.48	7	29.64	7.1073
23	23	499	869	903.436	46	19.6399	7.5211
24	9	207	349	267.61	17	15.7418	7.6147
25	5	168	322	118.498	6	19.7496	8.2887
26	4	177	394	201.723	5	40.3446	8.1657
27	1	35	62	7.8284	1	7.8284	12.9591
28	8	232	483	506.031	22	23.0014	8.1991
29	39	777	1166	1686.33	88	19.1629	8.6448
30	6	119	213	157.468	10	15.7468	7.2164
31	10	227	419	323.622	15	21.5748	8.0692
32	6	145	428	660.345	36	18.3429	10.4527

Normalised per lacuna

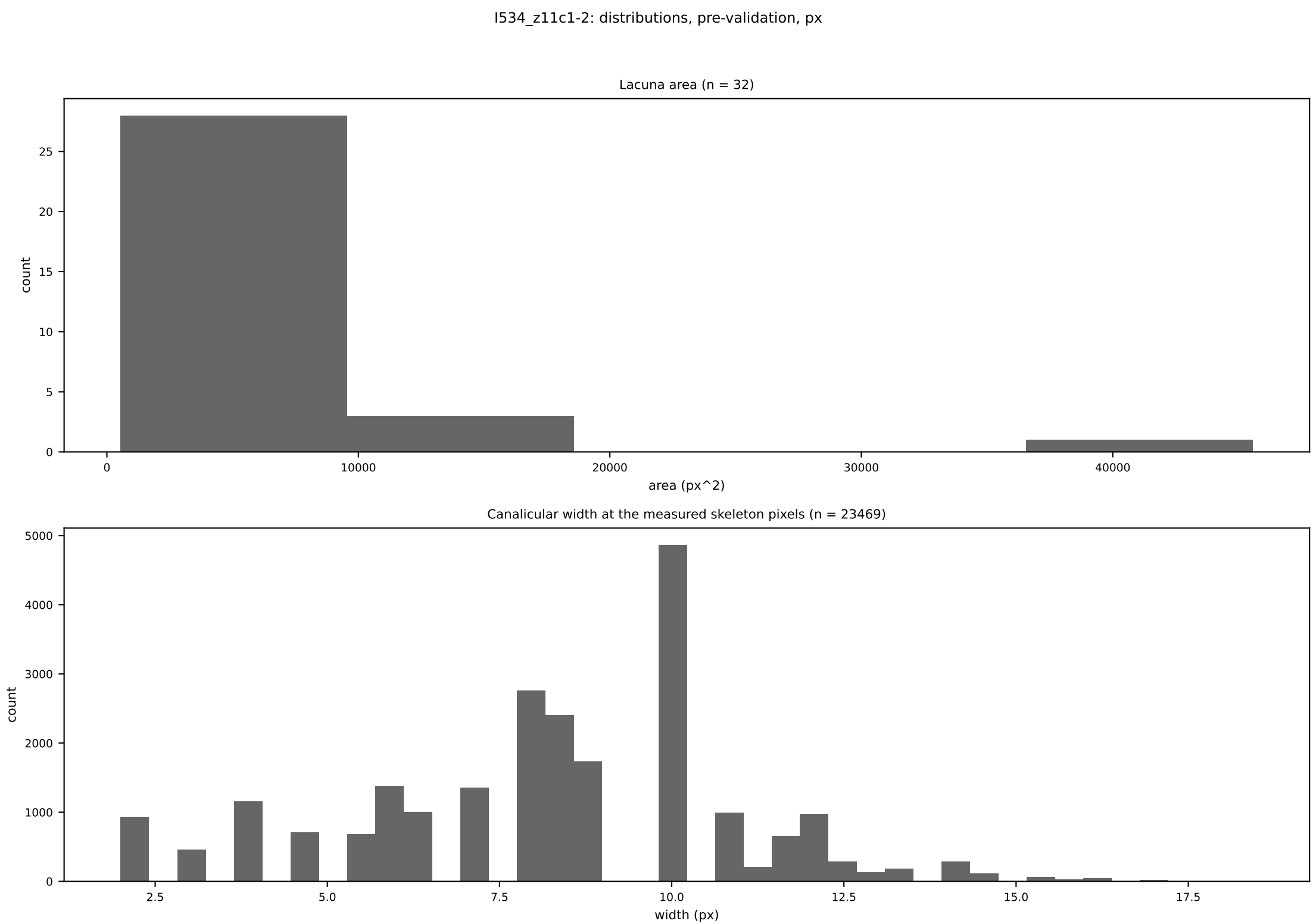
lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	494.351	8231	11715	0.7421	0.6708	0.0409428	0.036449	3.0343
2	282.291	4862	7142	0.9455	0.8134	0.0563554	0.050126	4.2509
3	1255.09	18964	24340	0.693	0.5601	0.0264712	0.0253492	1.9122
4	253.279	7100	12962	1	0.9073	0.0412676	0.044129	3.9482
5	270.635	8169	16049	0.8695	0.7737	0.0430897	0.0371986	3.3255
6	236.936	5018	6853	1	1	0.0478278	0.0468408	2.9544
7	341.505	10661	17446	1	1	0.0405215	0.0363407	4.3923
8	176.167	7690	19162	1	1	0.0418726	0.0416971	4.5412
9	1221.79	20165	28289	0.8577	0.7956	0.0255393	0.0278554	1.4732

10	199.782	5730	14247	0.6804	0.6335	0.0270506	0.0251281	3.0033
11	548.257	17191	33534	1	1	0.038334	0.0368283	4.7423
12	136.125	6658	12335	1	1	0.0306398	0.0316174	5.1423
13	297.706	11030	23297	1	1	0.0453309	0.0452848	5.3744
14	305.605	10599	17028	1	1	0.0321729	0.0386422	3.5994
15	191.167	7042	12138	0.9189	0.7962	0.0254189	0.0193607	3.6617
16	238.451	9266	19320	1	1	0.0388517	0.0345756	5.0325
17	122.953	5564	11136	1	1	0.0278577	0.0231681	4.0666
18	490.399	11727	20563	0.8061	0.7376	0.0299309	0.0247046	3.8744
19	378.475	12733	27257	1	1	0.0302364	0.0316249	5.0201
20	373.546	12922	30033	1	1	0.0315741	0.0342956	3.7479
21	290.877	7526	16187	1	1	0.0368057	0.0361401	5.1568
22	233.38	6410	12080	1	1	0.026053	0.0280629	3.8564
23	555.955	15261	25089	0.9401	0.8614	0.0326977	0.0346367	4.137
24	383.889	12892	25468	1	1	0.0160565	0.0137035	2.3444
25	97.9828	5438	10968	1	1	0.0308937	0.0293581	5.1029
26	115.539	5958	13707	1	1	0.029708	0.0287444	3.462
27	139.054	4710	7157	1	1	0.007431	0.00866285	0.7191
28	222.693	8786	20073	1	0.9917	0.0264056	0.0240622	3.5924
29	2556.07	33726	52704	0.6475	0.572	0.0230386	0.0221236	1.5258
30	196.066	6012	12715	1	1	0.0197937	0.0167519	3.0602
31	446.416	13647	28597	1	1	0.0166337	0.0146519	2.2401
32	206.752	4961	9598	0.6755	0.6216	0.029228	0.0445926	2.902

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	334	421	381.806	480.818	9	8	4
2	271	315	309.35	406.605	9	6	3
3	438	497	563.387	698.583	16	12	5
4	201	348	331.836	643.073	5	6	12
5	348	545	392.836	660.688	6	9	6
6	151	151	275.522	367.291	5	4	4
7	413	591	496.446	725.441	11	11	12
8	284	620	367.563	917.051	6	11	10
9	465	540	581.902	895.394	12	14	9
10	140	252	165.669	393.836	3	4	4
11	639	1187	726.487	1384.67	15	15	17
12	196	285	235.38	443.919	5	6	4
13	486	1025	568.688	1201.18	10	16	18
14	327	612	395.505	761.553	8	9	12
15	153	156	207.794	271.321	5	2	3
16	340	593	412.605	776.039	6	7	9
17	145	178	175.297	287.551	4	4	4
18	327	352	403.576	592.541	7	3	5
19	376	722	438.233	975.264	11	10	11
20	395	767	467.961	1158.28	8	11	12
21	232	384	316.179	665.186	3	6	7
22	153	183	181.853	382.019	6	3	3
23	451	748	578.399	1005.9	13	14	13
24	199	281	226.853	381.007	5	4	7
25	149	248	184.054	363.978	4	6	4
26	115	165	202.995	451.789	4	7	7
27	10	10	37.799	67.6985	0	1	2
28	232	429	267.794	549.902	6	6	7
29	716	1035	866.323	1308.97	31	16	9
30	119	159	135.397	236.238	2	2	3
31	181	252	251.137	462.576	7	7	9

32	139	405	158.468	481.161	5	4	4


I534_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 32)
count
area (px^2)
0
5
10
15
20
25
0
10000
20000
30000
40000
Canalicular width at the measured skeleton pixels (n = 23469)
count
width (px)
0
1000
2000
3000
4000
5000
2.5
5.0
7.5
10.0
12.5
15.0
17.5